INTRODUCTION

In this Learning Module, we present the various methods for obtaining information on a population (all members of a specified group) through samples (part of the population). The information on a population that we seek usually concerns the value of a **parameter**, a quantity computed from or used to describe a population of data. In Lesson 1 we introduce sampling, which we use a sample to estimate a parameter; we make use of sample statistics. A statistic is a quantity computed from or used to describe a sample of data.

Supposing that a sample is representative of the underlying population, how can the analyst assess the sampling error in estimating a population parameter? In Lesson 2, the Central Limit Theorem helps us understand the sampling distribution of the sample mean in many of the estimation problems we face. This provides guidance on how closely a sample mean can be expected to match its underlying population mean, allowing an analyst to use the sampling distribution to assess the accuracy of the sample and test hypotheses about the underlying parameter. Lesson 3 covers various resampling approaches.

Learning Module Overview

- Of the two types of sampling methods, probability sampling includes simple random sampling and stratified random sampling, and non-probability sampling includes convenience sampling and judgmental sampling. Probability sampling involves equal chance of sample selection, while non-probability sampling has a significant risk of being non-representative.
- Sampling error is the difference between the observed value of a statistic and the quantity it is intended to estimate as a result of using subsets of the population.
- Non-probability sampling methods rely not on a fixed selection process but instead on a researcher's sample selection capabilities. Its advantages include quick and low-cost data collection, and can apply expert judgment for efficient sample selection.
- The Central Limit Theorem is defined as follows: Given a population described by any probability distribution with mean μ and finite variance σ^2, the sampling distribution of the sample mean $\overline{X}$ computed from random samples of size n from this population will be approximately normal with mean μ (the population mean) and variance σ^2/n (the population variance divided by n) when the sample size n is large.
- The standard error of the sample mean is an important quantity in applying the central limit theorem in practice. It is typically estimated using the square root of the sample variance, calculated as follows:

$$s^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}.$$

- The central limit theorem shows that when sampling from any distribution, the sample mean distribution will have these two properties when the sample size is large: (1) the distribution of the sample mean $\overline{X}$ will be approximately normal, and (2) the mean of the distribution of $\overline{X}$ will be equal to the mean of the population from which samples are drawn.
- Bootstrap, a popular resampling method which repeatedly draws samples of the same size as the original sample, uses computer simulation for statistical inference without using an analytical formula such as a z-statistic or t-statistic. It can be used as a simple but powerful method for any complicated estimators such as the standard error of a sample mean.
- Bootstrap has potential advantages in accuracy. Given these advantages, it can be applied widely in finance, such as for historical simulations in asset allocation or in gauging an investment strategy's performance against a benchmark.
- Jackknife is another resampling technique with samples selected by taking the original observed data sample and leaving out one observation at a time from the set (and not replacing it). Jackknife is often used to reduce the bias of an estimator, and other applications include finding the standard error and confidence interval of an estimator.

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